

Solution exercise set 3

Solutions to the problems from the book are found at the end.

Problem 1:

For $i = 1, \dots, 10$ let

$$I_j = \begin{cases} 0, & \text{with probability } 1 - p_j \\ 1, & \text{with probability } p_j \end{cases}$$

where $I_j = 1$ means that component j works. Then

a) $P(I_1 = 1 \cap I_2 = 1) = P(I_1 = 1)P(I_2 = 1) = p_1 p_2 = 0.7 \cdot 0.95 = \underline{0.665}$.

b) $P(I_1 = 1 \cup I_2 = 1) = P(I_1 = 1) + P(I_2 = 1) - P(I_1 = 1)P(I_2 = 1) = p_1 + p_2 - p_1 p_2 = \underline{0.985}$.

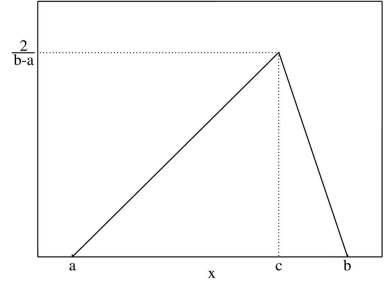
c) $P\{(I_1 = 1 \cap I_2 = 1) \cap (I_3 = 1 \cup I_4 = 1)\}$
 $= P\{(I_1 = 1 \cap I_2 = 1 \cap I_3 = 1) \cup (I_1 = 1 \cap I_2 = 1 \cap I_4 = 1)\}$
 $= p_1 p_2 p_3 + p_1 p_2 p_4 - p_1 p_2 p_3 p_4 = \underline{0.663}$.

d) this probability by simulation! See the R-code file for details.

Problem 2:

If a random variable Y has a triangle distribution with minimum a , maximum b and mode c ($a < c < b$), we have from the lectures that the density of Y is

$$f_Y(y) = \begin{cases} \frac{2(y-a)}{(b-a)(c-a)}, & \text{for } a \leq y \leq c \\ \frac{2(b-y)}{(b-c)(b-a)}, & \text{for } c \leq y \leq b \\ 0, & \text{else} \end{cases}$$



We will simulate a lot from this distribution in this exercise so we first consider how to simulate from this distribution in general.

In the special case of a symmetric triangle distribution (i.e. $c = (a + b)/2$) there is a result that if $Y = U_1 + U_2$, where U_1 and U_2 are independent $U[\frac{a}{2}, \frac{b}{2}]$, then Y has this symmetric triangle distribution. Therefore for the special case of symmetric triangle distributions we can easily simulate the distribution by adding to independent $U[\frac{a}{2}, \frac{b}{2}]$ variables. Otherwise we have to use some other method, for instance acceptance-rejection.

Simulation by the acceptance-rejection (AR) method:

Let X have a triangle distribution with minimum a , maximum b and mode c ($a < c < b$); denote it's density by $f(x)$, c.f. above.

- Let $Y \sim U[a, b]$, i.e. Y has density function $g(y) = 1/(b - a)$ for $a \leq y \leq b$.
- We have that $\frac{f(y)}{g(y)} = \frac{f(y)}{1/(b - a)} \leq \frac{2/(b - a)}{1/(b - a)} = 2$, c.f. above.
- Thus, algorithm for AR simulation:
 1. Generate Y from the distribution with density g .
 2. Generate U from the $U[0, 1]$ distribution.
 3. If $U \leq \frac{f(Y)}{2 \cdot g(Y)}$, Y is accepted, else rejected.

The expression $\frac{f(Y)}{2 \cdot g(Y)}$ in step 3 above:

$$\frac{f(Y)}{2 \cdot g(Y)} = \begin{cases} \frac{2(y-a)}{(b-a)(c-a)} \cdot \frac{(b-a)}{2} = \frac{(y-a)}{(c-a)}, & \text{for } a \leq y \leq c \\ -\frac{2(y-b)}{(b-c)(b-a)} \cdot \frac{(b-a)}{2} = \frac{-(y-b)}{(b-c)} = \frac{(b-y)}{(b-c)}, & \text{for } c \leq y \leq b \end{cases}$$

a) Conditional on there being a find, the size of the resources is distributed as a triangle distribution with minimum $a = 2$, maximum $b = 10$ and mode $c = 6$ and therefore has the density:

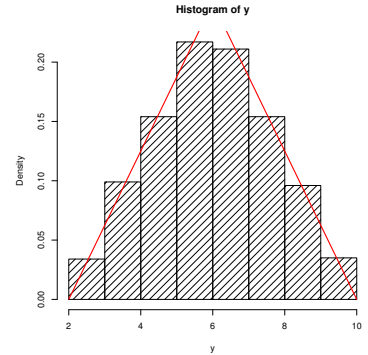
$$f_Y(y) = \begin{cases} \frac{2(y-2)}{(10-2)(6-2)} = \frac{2(y-2)}{32}, & \text{for } 2 \leq y \leq 6 \\ \frac{2(10-y)}{(10-6)(10-2)} = -\frac{2(y-10)}{32}, & \text{for } 6 \leq y \leq 10 \\ 0, & \text{else} \end{cases}$$

Histogram of 1000 simulated values of Y together with true density (in red).

This is the **conditional distribution of the resource size** given a finding.

Simulation estimate of $P(Y \geq 8)$: 0.131

This is the estimated probability of resource size ≥ 8 conditional on there being a finding.



Let X be the size of the resources (unconditional). Then we have:

$$X = \begin{cases} 0, & \text{with probability 0.6} \\ Y, & \text{with probability 0.4} \end{cases}$$

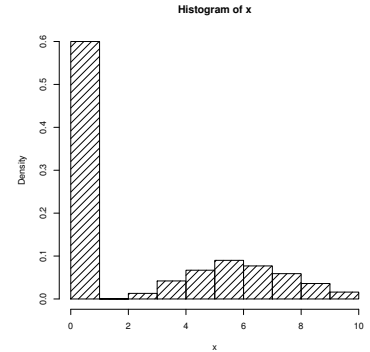
where Y has the triangle distribution described above.

Histogram of 1000 simulated values of X .

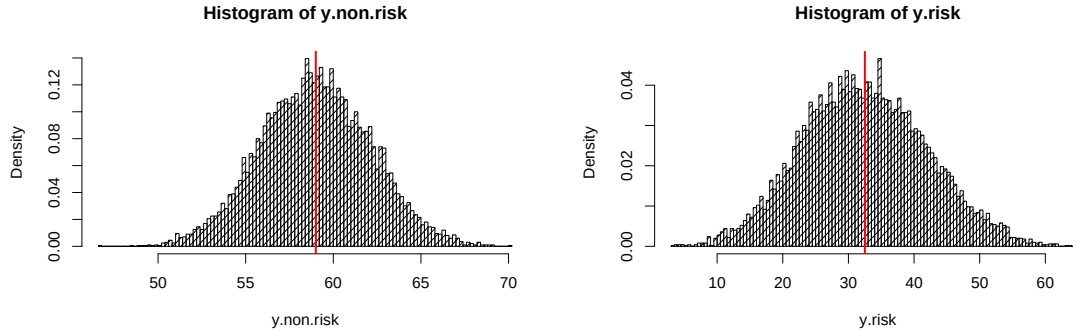
This is the **unconditional distribution of the resource size**.

Simulation estimate of $P(X \geq 8)$: 0.052

This is the estimated unconditional probability of resource size ≥ 8 .



b) The plots below show histograms for 10 000 simulations of the total resources for the not risk-weighted (left) case and the risk-weighted case (right). The estimated expectations are respectively 59.0 and 32.5.



c) Recall that for the triangular distribution $E(X) = (a + b + c)/3$. and that $E(R) = E(\sum_{i=1}^{10} X_i) = \sum_{i=1}^{10} E(X_i)$.

For the not risk-weighted case the $E(X_i)$ are found directly from the formula for the triangle distribution, $E(X_i) = (a_i + b_i + c_i)/3$. For the risk-weighted case we do again have that

$$X_i = \begin{cases} 0, & \text{with probability } 1 - p_i \\ Y_i, & \text{with probability } p_i \end{cases},$$

where Y has the triangle distribution. Thus in this case $E(X_i) = 0 \cdot (1 - p_i) + E(Y_i) \cdot p_i = p_i(a_i + b_i + c_i)/3$.

The calculations are done in R (see the R-code file) and give respectively $E(R) = \underline{59}$ in the not risk-weighted case and $E(R) = \underline{32.7}$ in the risk-weighted case.

d) See the R-code file. Using the formula from the lecture notes we need more than $4\sigma^2/e^2$ replications to be 95% certain to not be more than an error e from the true value. The calculations are done in the R-code file. We seem to need around 990 replications in the not risk-weighted case and around 8900 replications in the risk-weighted case.

Problem 3:

a) The covariance matrix for the bivariate normal variable is $\Sigma = \begin{bmatrix} \sigma_{XX} & \sigma_{XY} \\ \sigma_{YX} & \sigma_{YY} \end{bmatrix}$ where $\sigma_{XX} = \text{Var}(X) = 1$, $\sigma_{YY} = \text{Var}(Y) = 3$ and $\sigma_{XY} = \sigma_{YX} = \text{Cov}(X, Y)$. Recall from the lecture that

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

which implies

$$\text{Cov}(X, Y) = \rho(X, Y)\sqrt{\text{Var}(X)\text{Var}(Y)} = 0.7 \cdot \sqrt{1 \cdot 3} = 1.21$$

and thus:

$$\Sigma = \begin{bmatrix} 1 & 1.21 \\ 1.21 & 3 \end{bmatrix}$$

b) See the R-code file.

c) See the R-code file.

d) To find these probabilities analytically we use the fact that linear combination of normally distributed variables also have a normal distribution. For $V = X + Y$ we then further need to calculate $E(V)$ and $\text{Var}(V)$.

In both cases $E(V) = E(X) + E(Y) = 1 + 1 = 2$.

When $\rho(X, Y) = 0$ we get that $\text{Var}(V) = \text{Var}(X) + \text{Var}(Y) = 1 + 3 = 4$, while when $\rho(X, Y) = 0.7$ and thus $\text{Cov}(X, Y) = 1.21$ as shown in point a) we get that $\text{Var}(V) = \text{Var}(X) + \text{Var}(Y) + 2 \cdot 1 \cdot 1 \cdot \text{Cov}(X, Y) = 1 + 3 + 2 \cdot 1.21 = 6.42$ (see results for expectation and variance for linear combinations in the lecture notes (chapter 2, part 1)).

Then:

$$\begin{aligned} P(V \geq 3) &\stackrel{\rho=0.7}{=} 1 - P\left(\frac{V - E(V)}{\sqrt{\text{Var}(V)}} < \frac{3 - E(V)}{\sqrt{\text{Var}(V)}}\right) = 1 - P\left(Z < \frac{3 - 2}{\sqrt{6.42}}\right) \\ &= 1 - P(Z < 0.39) = 1 - \text{pnorm}(0.39) = 1 - 0.65 = \underline{\underline{0.35}} \\ P(V \geq 3) &\stackrel{\rho=0}{=} 1 - P\left(\frac{V - E(V)}{\sqrt{\text{Var}(V)}} < \frac{3 - E(V)}{\sqrt{\text{Var}(V)}}\right) = 1 - P\left(Z < \frac{3 - 2}{\sqrt{4}}\right) \\ &= 1 - P(Z < 0.5) = 1 - \text{pnorm}(0.5) = 1 - 0.69 = \underline{\underline{0.31}} \end{aligned}$$

Here `pnorm` is the R function to compute cumulative standard normal probabilities.

Exercises from the book:

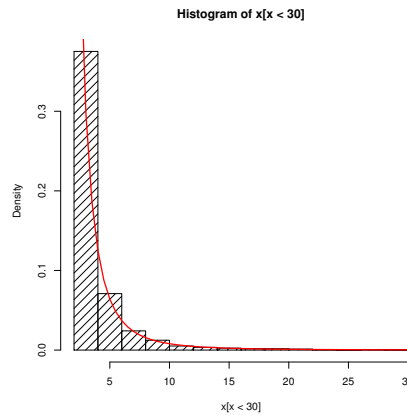
Rizzo, exercise 3.3:

Pareto(a, b) distribution, cumulative distribution function: $F(x) = 1 - \left(\frac{b}{x}\right)^a$, for $0 < b \leq x$ and $a > 0$.

Density: $f(x) = F'(x) = -a\left(\frac{b}{x}\right)^{a-1} \left(\frac{-b}{x^2}\right) = \frac{ab^a}{x^{a+1}}$.

$$U = F(X) \Rightarrow X = \frac{b}{\sqrt[a]{1-U}}$$

`n.replic = 1000` replications from the Pareto(2, 2) distribution compared with theoretical density (red):

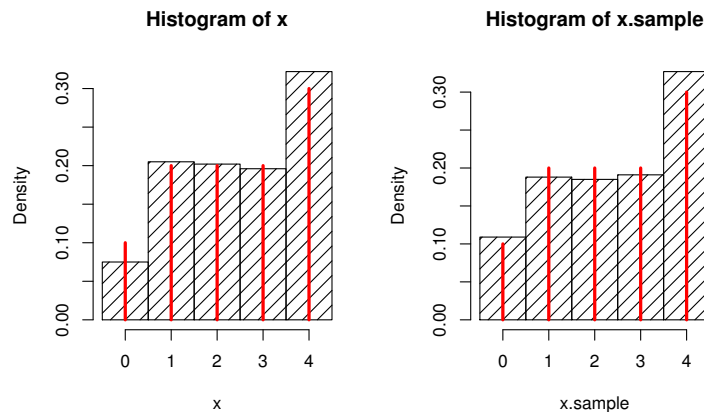


The histogram is based only on values < 30 . There were 54 simulated values ≥ 30 .

See the R-code file for implementation and further details.

Rizzo, exercise 3.5:

See the R-code file. Relative frequency histogram with the true probabilities in red for comparison:



Rizzo, exercise 3.7:

Beta(α, β) density: $f(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$, $0 \leq x \leq 1$.

Following Rizzo pp. 56, 57, we take g as the uniform (0,1) density. We need to find maximum of $f(x)$, $0 \leq x \leq 1$, to find the value of c for which $\frac{f(x)}{g(x)} = \frac{f(x)}{1} = f(x) \leq c$ for all $0 \leq x \leq 1$.

We restrict attention to the case $\alpha > 1$ and $\beta > 1$ for which the beta density is bounded.

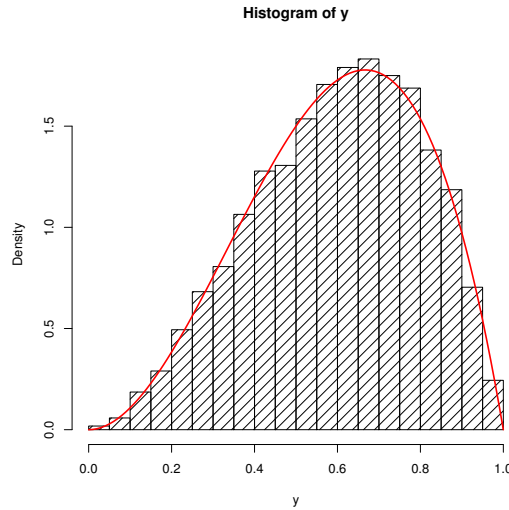
Derivative equal to zero to find extremes of $f(x)$:

$$\begin{aligned} f'(x) &= \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \left[(\alpha - 1)x^{\alpha-2}(1-x)^{\beta-1} - (\beta - 1)x^{\alpha-1}(1-x)^{\beta-2} \right] = 0 \\ (\alpha - 1)x^{\alpha-2}(1-x)^{\beta-1} &= (\beta - 1)x^{\alpha-1}(1-x)^{\beta-2} \\ (\alpha - 1)(1-x) &= (\beta - 1)x \\ (\alpha - 1) &= (\beta - 1)x + (\alpha - 1)x \\ \Rightarrow x_{max} &= \frac{\alpha - 1}{\alpha - 1 + \beta - 1} = \frac{\alpha - 1}{\alpha + \beta - 2} \end{aligned}$$

Thus $c = f(\frac{\alpha-1}{\alpha+\beta-2})$ and further we have: $\frac{f(x)}{cg(x)} = \frac{f(x)}{f(\frac{\alpha-1}{\alpha+\beta-2})}$.

Thus, for $X \sim U(0, 1)$ and $U \sim U(0, 1)$ independent, accept X if $U < \frac{f(X)}{f(\frac{\alpha-1}{\alpha+\beta-2})}$, reject otherwise.

See the R-code file and the plot below.



Rizzo, exercise 3.17:

See the R-code file.